CAMPUS CÁMARA COMERCIO SEVILLA
FP SUPERIOR - GRADOS UNIVERSITARIOS - MÁSTERES

Cámara Sevilla

AEE. Bitácora III. Conexiones Empresariales
Taller Técnico: Operación Escudo

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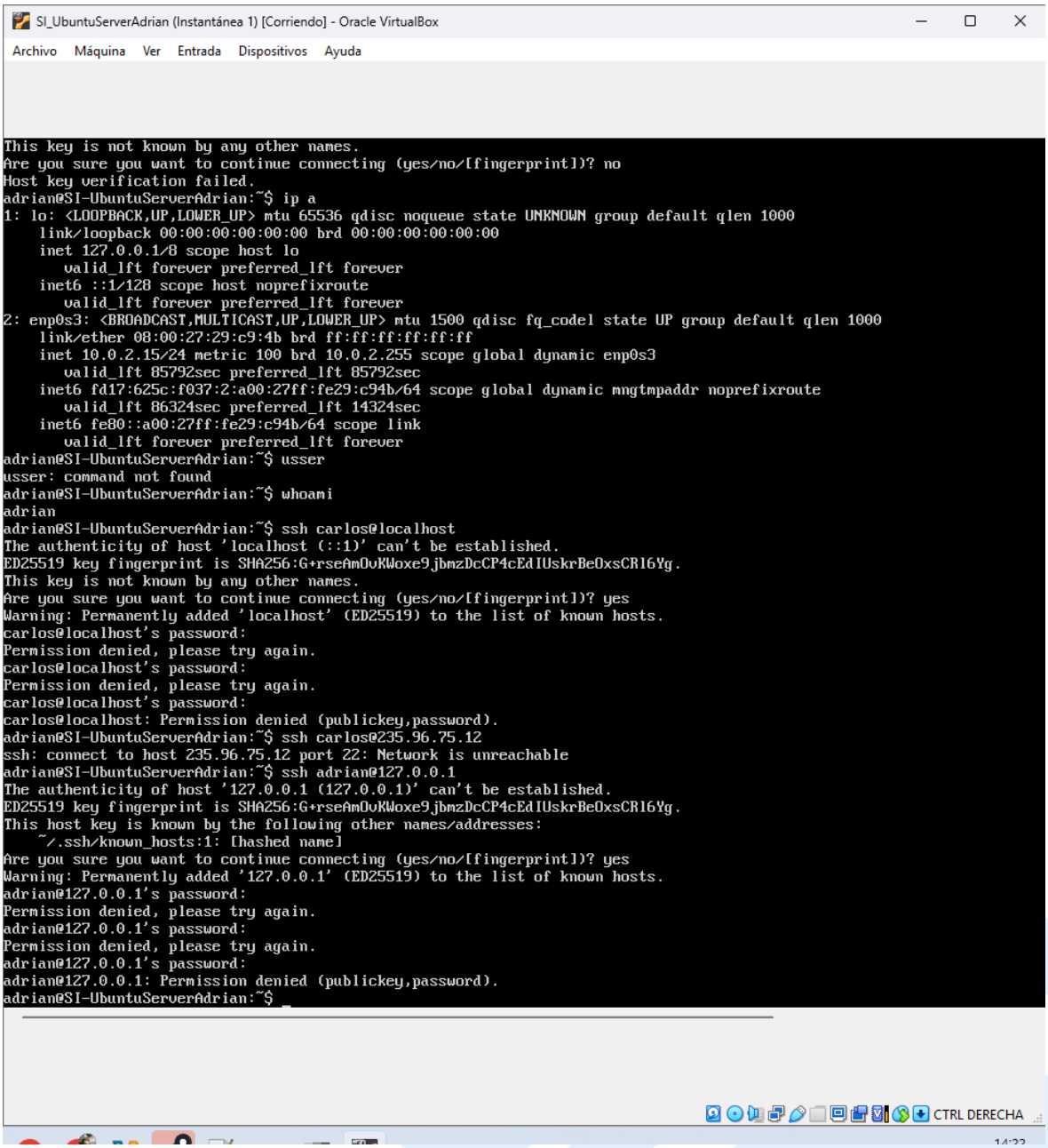
Asignatura: Sistemas Informáticos

Unión Europea

Me bloquea el usuario después de 5 errores



1. **Análisis Forense Manual**.

```
grep: password@: No such file or directory
/var/log/auth.log:2026-04-17T12:24:48.341651+00:00 SI-UbuntuServerAdrian sudo:   adrian : TTY=tty1 ; PWD=/home/adrian ;
d password@ /var/log/auth.log
adrian@SI-UbuntuServerAdrian:~$ sudo grep  "Failed password" /var/log/auth.log |tail -n 5
2026-04-17T12:20:31.169272+00:00 SI-UbuntuServerAdrian sshd[3502]: Failed password for invalid user carlos from ::1 port
2026-04-17T12:22:20.629786+00:00 SI-UbuntuServerAdrian sshd[3514]: Failed password for adrian from 127.0.0.1 port 48854
2026-04-17T12:22:34.428775+00:00 SI-UbuntuServerAdrian sshd[3514]: message repeated 2 times: [ Failed password for adria
2026-04-17T12:24:48.341651+00:00 SI-UbuntuServerAdrian sudo:   adrian : TTY=tty1 ; PWD=/home/adrian ; USER=root ; COMMAN
og/auth.log
2026-04-17T12:25:23.187688+00:00 SI-UbuntuServerAdrian sudo:   adrian : TTY=tty1 ; PWD=/home/adrian ; USER=root ; COMMAN
og/auth.log
adrian@SI-UbuntuServerAdrian:~$ _
```

Creamos el txt para luego mostrar los errores de acceso por pantalla

```
  GNU nano 7.2                                        /home/adri/scripts/check_intruders.sh
#!/bin/bash
LOG_SOURCE="/var/log/auth.log"
REPORT_DEST="$HOME/alertas.txt"
echo "==================================" > $REPORT_DEST
echo "INFORME DE SEGURIDAD - $(date)"  >> $REPORT_DEST
echo "==================================" >> $REPORT_DEST
echo "" >> $REPORT_DEST
echo "DETALLE DE INTENTOS FALLIDOS: " >> $REPORT_DEST

grep -i "Failed password" $LOG_SOURCE | awk '{print " Fecha: " $1 " | Usuario: " $9 " | IP: "  $11}' >> $REPORT_DEST

TOTAL=$(grep -i -c "failed password" $LOG_SOURCE)

echo  >> $REPORT_DEST
echo "-------------------------------------------" >> $REPORT_DEST
echo "RESUMEN FINAL: Se han detectado $TOTAL intentos" >> $REPORT_DEST
echo "-------------------------------------------" >> $REPORT_DEST

echo "Analisis completado. El reporte se ha guardado en:  $REPORT_DEST"

                                                  [ Read 19 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^- Go To Line  M-E Redo     M-6 Copy
<                                                                                                  >
```

Podemos observar que gracias a ejecutar el txt anterior nos muestra los tres errores a la hora de querer acceder al servidor

```
adri@adri:~$ sudo ~/scripts/check_intruders.sh
Analisis completado. El reporte se ha guardado en:  /root/alertas.txt
adri@adri:~$ sudo cat /root/alertas.txt
====================================
INFORME DE SEGURIDAD - Fri Apr 17 06:41:49 PM UTC 2026
====================================

DETALLE DE INTENTOS FALLIDOS:
 Fecha: 2026-04-17T17:55:12.863815+00:00 | Usuario: carlos | IP: 127.0.0.1
 Fecha: 2026-04-17T17:55:19.159606+00:00 | Usuario: carlos | IP: 127.0.0.1
 Fecha: 2026-04-17T17:55:29.740911+00:00 | Usuario: carlos | IP: 127.0.0.1

-------------------------------------------
RESUMEN FINAL: Se han detectado 3 intentos
-------------------------------------------
adri@adri:~$ _
```

Reto B: Configuración del "Escudo" Perimetral con UFW
Consigo abrir los puertos 22.

```
adri@adri:~$ sudo cat /root/alertas.txt
====================================
INFORME DE SEGURIDAD - Fri Apr 17 06:41:49 PM UTC 2026
====================================

DETALLE DE INTENTOS FALLIDOS:
 Fecha: 2026-04-17T17:55:12.863815+00:00 | Usuario: carlos | IP: 127.0.0.1
 Fecha: 2026-04-17T17:55:19.159606+00:00 | Usuario: carlos | IP: 127.0.0.1
 Fecha: 2026-04-17T17:55:29.740911+00:00 | Usuario: carlos | IP: 127.0.0.1

-------------------------------------------
RESUMEN FINAL: Se han detectado 3 intentos
-------------------------------------------
adri@adri:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
adri@adri:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
adri@adri:~$ sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
adri@adri:~$ _
```

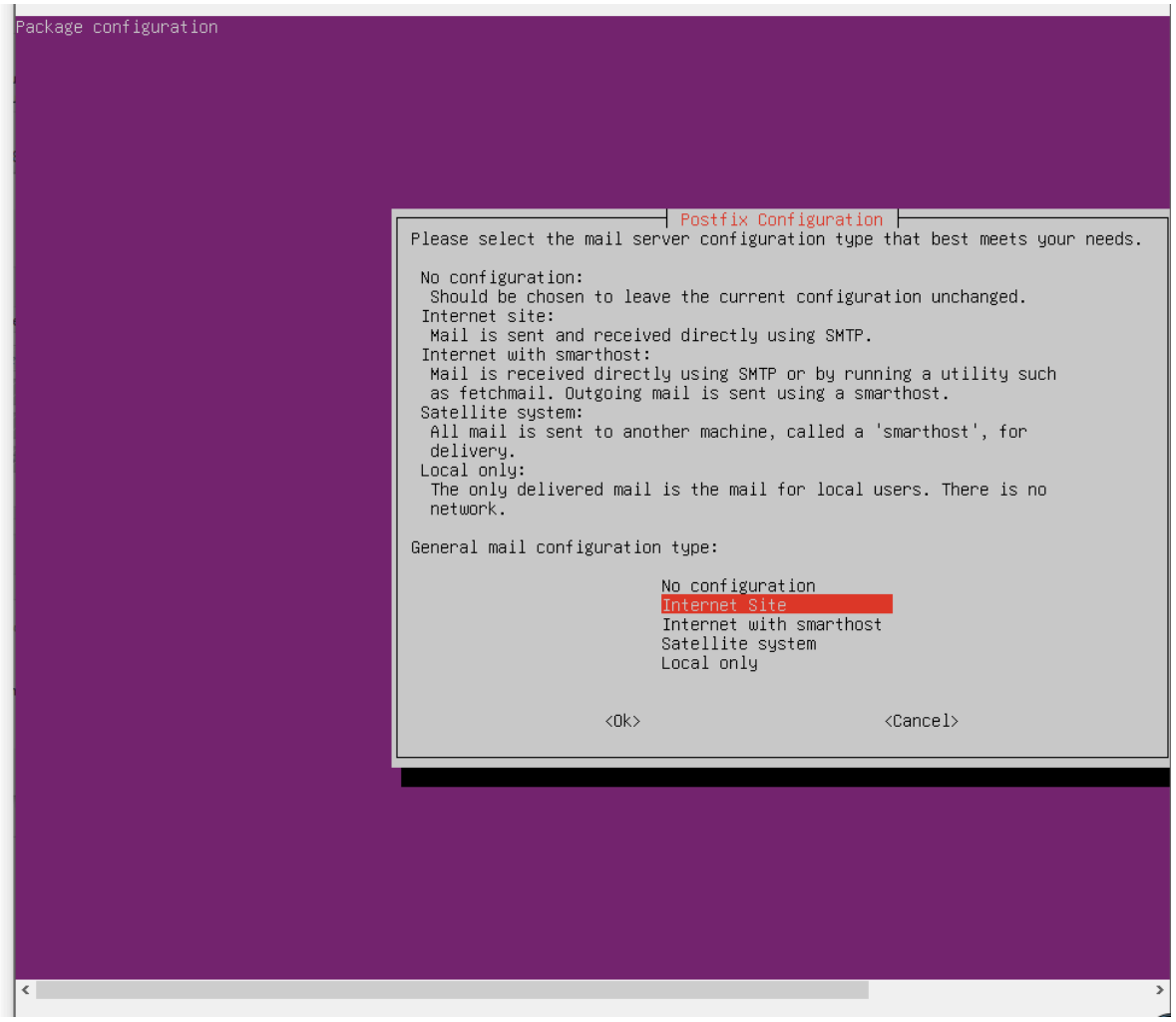
Modificó el ACCEPT PO DROP

```
GNU nano 7.2                                                    /etc/ufw/before.rules
#
# rules.before
#
# Rules that should be run before the ufw command line added rules. Custom
# rules should be added to one of these chains:
#   ufw-before-input
#   ufw-before-output
#   ufw-before-forward
#

# Don't delete these required lines, otherwise there will be errors
*filter
:ufw-before-input - [0:0]
:ufw-before-output - [0:0]
:ufw-before-forward - [0:0]
:ufw-not-local - [0:0]
# End required lines


# allow all on loopback
-A ufw-before-input -i lo -j ACCEPT
-A ufw-before-output -o lo -j ACCEPT

# quickly process packets for which we already have a connection
-A ufw-before-input -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A ufw-before-output -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A ufw-before-forward -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT

# drop INVALID packets (logs these in loglevel medium and higher)
-A ufw-before-input -m conntrack --ctstate INVALID -j ufw-logging-deny
-A ufw-before-input -m conntrack --ctstate INVALID -j DROP

# ok icmp codes for INPUT
-A ufw-before-input -p icmp --icmp-type destination-unreachable -j ACCEPT
-A ufw-before-input -p icmp --icmp-type time-exceeded -j ACCEPT
-A ufw-before-input -p icmp --icmp-type parameter-problem -j ACCEPT
-A ufw-before-input -p icmp --icmp-type echo-request -j DROP

# ok icmp code for FORWARD
-A ufw-before-forward -p icmp --icmp-type destination-unreachable -j ACCEPT
-A ufw-before-forward -p icmp --icmp-type time-exceeded -j ACCEPT
-A ufw-before-forward -p icmp --icmp-type parameter-problem -j ACCEPT
-A ufw-before-forward -p icmp --icmp-type echo-request -j ACCEPT

# allow dhcp client to work
-A ufw-before-input -p udp --sport 67 --dport 68 -j ACCEPT


adri@adri:~$ _
```

Como se muestra en la siguiente captura se muestra el puerto 22 abierto, la política deny (incoming) activada y el cambio a DROP en el archivo de reglas.

```
-A ufw-before-input -i lo -j ACCEPT
-A ufw-before-output -o lo -j ACCEPT

# quickly process packets for which we already have a connection
-A ufw-before-input -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A ufw-before-output -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A ufw-before-forward -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT

# drop INVALID packets (logs these in loglevel medium and higher)
-A ufw-before-input -m conntrack --ctstate INVALID -j ufw-logging-deny
-A ufw-before-input -m conntrack --ctstate INVALID -j DROP

# ok icmp codes for INPUT
-A ufw-before-input -p icmp --icmp-type destination-unreachable -j ACCEPT
-A ufw-before-input -p icmp --icmp-type time-exceeded -j ACCEPT
-A ufw-before-input -p icmp --icmp-type parameter-problem -j ACCEPT
-A ufw-before-input -p icmp --icmp-type echo-request -j DROP

# ok icmp code for FORWARD
-A ufw-before-forward -p icmp --icmp-type destination-unreachable -j ACCEPT
-A ufw-before-forward -p icmp --icmp-type time-exceeded -j ACCEPT
-A ufw-before-forward -p icmp --icmp-type parameter-problem -j ACCEPT
-A ufw-before-forward -p icmp --icmp-type echo-request -j ACCEPT

# allow dhcp client to work
-A ufw-before-input -p udp --sport 67 --dport 68 -j ACCEPT


adri@adri:~$ sudo ufw enable
Firewall is active and enabled on system startup
adri@adri:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)

adri@adri:~$ sudo ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     ALLOW IN    Anywhere
[ 2] 22/tcp (v6)                ALLOW IN    Anywhere (v6)

adri@adri:~$ _
```

Configurar el correo donde se enviarán los mensajes en caso de acceso denegado.

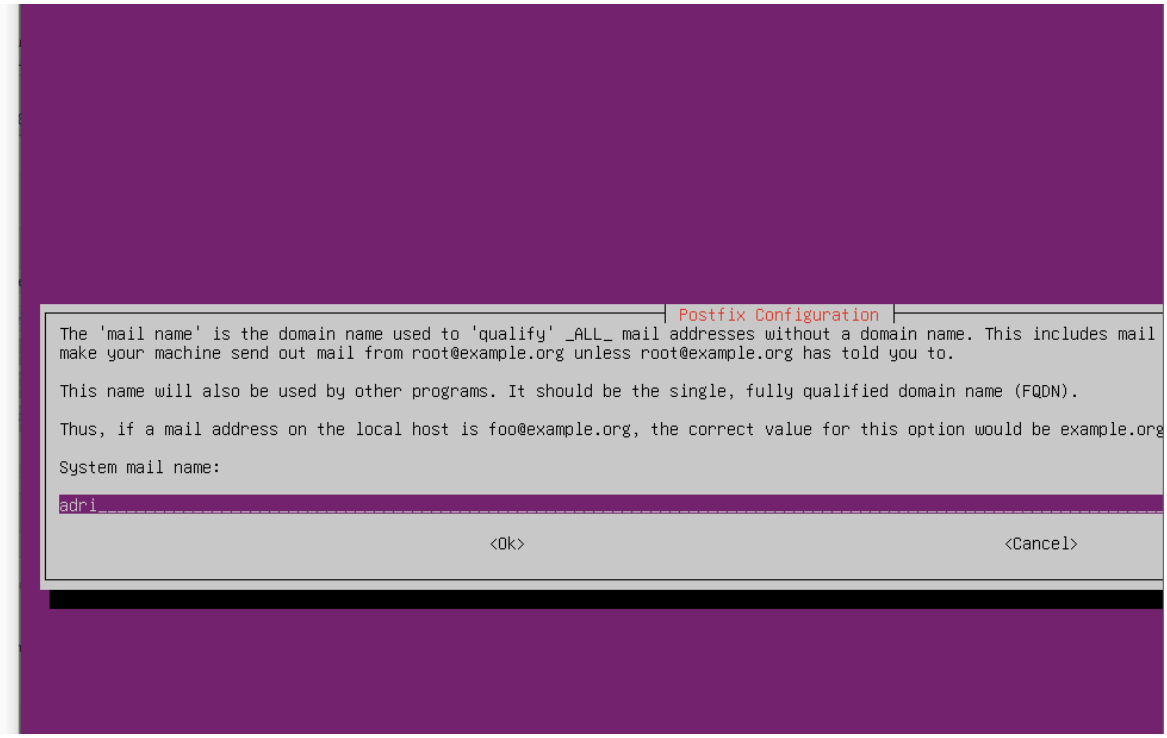


Inicializo y activo la base de datos.

ubuntu_server [Corriendo] - Oracle VirtualBox

Archivo Máquina Ver Entrada Dispositivos Ayuda

```
Creating config file /etc/aide/aide.conf.d/31_aide_pcscd with new version

Creating config file /etc/aide/aide.conf.d/31_aide_ippl with new version

Creating config file /etc/aide/aide.conf.d/31_aide_unbound with new version

Creating config file /etc/aide/aide.conf.d/31_aide_systemd-networkd with new version

Creating config file /etc/aide/aide.conf with new version

Creating config file /etc/aide/aide.settings.d/10_aide_sourceslist with new version

Creating config file /etc/aide/aide.settings.d/31_aide_trac_settings with new version

Creating config file /etc/aide/aide.settings.d/31_aide_svn-server_settings with new version

Creating config file /etc/aide/aide.settings.d/31_aide_torrus_settings with new version

Creating config file /etc/aide/aide.settings.d/31_aide_apt_settings with new version

Creating config file /etc/cron.daily/dailyaidecheck with new version
Creating group '_aide' with GID 988.
Creating user '_aide' (Advanced Intrusion Detection Environment) with UID 988 and GID 988.
Created symlink /etc/systemd/system/timers.target.wants/dailyaidecheck.timer → /usr/lib/systemd/system/dailyaidecheck.ti
dailyaidecheck.service is a disabled or a static unit, not starting it.
dailyaidecheck.service is a disabled or a static unit, not starting it.
Setting up bsd-mailx (8.1.2-0.20220412cvs-1build1) ...
update-alternatives: using /usr/bin/bsd-mailx to provide /usr/bin/mailx (mailx) in auto mode
Processing triggers for rsyslog (8.2312.0-3ubuntu9.1) ...
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
adri@adri:~$ sudo aideinit
Running aide --init...
adri@adri:~$ sudo cp /var/lib/aide/aide.db.new/ /var/lib/aide/aide.db
cp: cannot stat '/var/lib/aide/aide.db.new/': Not a directory
adri@adri:~$ sudo cp /var/lib/aide/aide.db.new /var/lib/aide/aide.db
adri@adri:~$
```



sudo aide --check --config /etc/aide/aide.conf | grep -i -A 20 "summary" a través de este comando se ve claramente que AIDE ha detectado **1 entrada añadida**

ubuntu_server [Corriendo] - Oracle VirtualBox

Archivo Máquina Ver Entrada Dispositivos Ayuda

```
---------------------------------------------------
The attributes of the (uncompressed) database(s):
---------------------------------------------------

/var/lib/aide/aide.db
 MD5       : PXKp0wF4pjQgzkovPbwDnQ==
 SHA1      : zmg9QJ0W00GcMIqpN8PmJjd6nOM=
 SHA256    : 0R+qayLQGneATOCeycjHH8d1URSRV29I
             29NFw1zPgpg=
 SHA512    : Pk3Ia4ghmMS32FHF7yzgEVr7D+E3nrqN
             BU2sII/IB/bGSK4cEkGFCMP9F6Y7LTTd
             i4y3UQNcZX9rWAh7dmo77g==
 RMD160    : VLg0t/Mg+3FH5tlZc7Prjmix/7A=
 TIGER     : yLfYx78bI3AcXi5bgezdgOvcPnUL4fE8
 CRC32     : M+nvAA==
 CRC32B    : lxkNog==
 HAVAL     : khKUzKkjxQuKPhyGFMNXbx7YOx0tZHH1
             qzfiJs/oSGs=
 WHIRLPOOL : dBm0RRc127IAlSfaoYGE2W4vB6wL/5YU
             3+fquVyvgOeMl7+rSLIBWdDjdFS1t93H
             kYY6X3l5CNG9roW3EdTGyQ==
 GOST      : TApDy4PRXiIfet+9dYAE8yMyucx2JH/S
             ntCu6MNyjrY=


End timestamp: 2026-04-17 19:08:22 +0000 (run time: 3m 12s)
adri@adri:~$ sudo aide --check --config /etc/aide/aide.conf | grep -A 20 "SUMMARY"
adri@adri:~$ sudo aide --check --config /etc/aide/aide.conf | grep -i -A 20 "summary"
Summary:
  Total number of entries:      100498
  Added entries:                1
  Removed entries:              0
  Changed entries:              3

---------------------------------------------------
Added entries:
---------------------------------------------------

f++++++++++++++++: /etc/test_robo

---------------------------------------------------
Changed entries:
---------------------------------------------------

f =.... mc..... . : /var/lib/systemd/timers/stamp-fwupd-refresh.timer
f >b... mc..H.. . : /var/log/aide/aideinit.log
f >b... mc..H.. . : /var/log/sysstat/sa17

---------------------------------------------------
adri@adri:~$ _
```

CTRL DERECHA